

Lesson 8: Newton's Second Law: with polar coordinates

Recall that Newton's second law is

$$\sum \vec{F} = m\vec{a}$$

In polar coordinates, this is recast as

$$\sum F_r = m(\ddot{r} - r\dot{\theta}^2) \quad \sum F_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta})$$

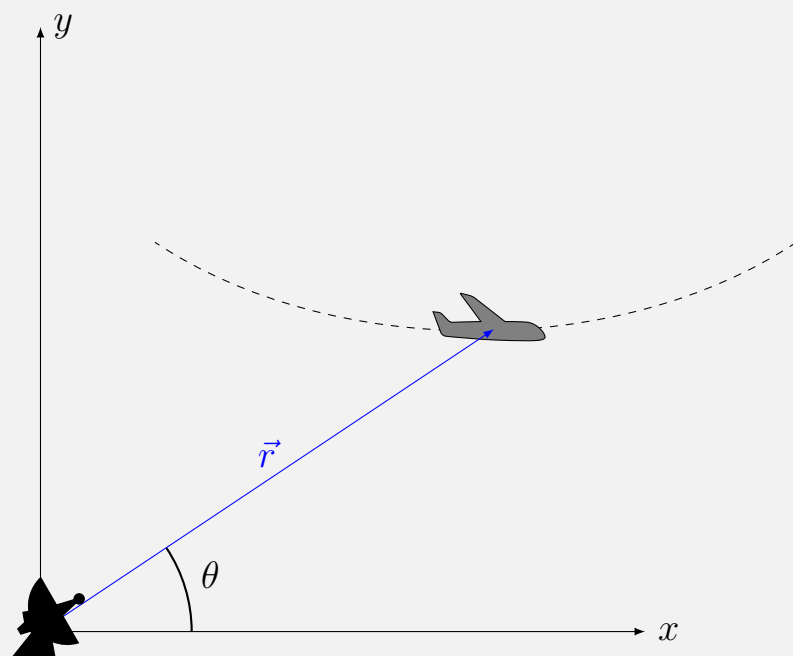
Sample Problem 8.1

A radar tracking station provides the following raw data at a given instant in time.

$r = 1000 \text{ m}$	$\theta = 36.87^\circ$
$\dot{r} = 120 \text{ m/s}$	$\dot{\theta} = -0.090 \text{ rad/s}$
$\ddot{r} = 34.8 \text{ m/s}^2$	$\ddot{\theta} = 0.0156 \text{ rad/s}^2$

Knowing that the airplane is at the bottom of its trajectory (*i.e.* the plane is horizontal) determined the force exerted on a 80 kg pilot by the seat.

8.1 – cont



$$F_x = 1997 \text{ N} \rightarrow$$

$$F_y = 1682 \text{ N} \uparrow$$

Sample Problems

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